AI & GENERATIVE AI TRACK / PROJECT 04 OF 05

LearnMate

PHASE 02 OF 08 | REQUIREMENT ANALYSIS

Project overview

LearnMate helps students work with scattered course materials through a REST API for uploads and AI-generated learning resources. The brief describes secure accounts, MongoDB storage, Gemini-powered content generation, and a dashboard-facing backend for saved study resources.

Project description

A study-support API for uploading learning materials and generating summaries, flashcards, quizzes, and personalized study plans.

Project status

Not stated in the source brief. Update this document when the team records an agreed status.

Project capabilities

- Study material uploads
- AI summaries and flashcards
- Practice quizzes
- Personalized study plans

Phase description

Turn the validated problem into functional, non-functional, data-flow, and technology requirements.

Project-specific phase focus

Define uploads, saved resources, access control, and the four AI-generated study formats.

Phase status and progress

Status: Not stated in the source material.
Progress: Not recorded in the source material.

Tasks and requirements

[ ] Map the customer journey and touchpoints
[ ] Diagram system data flows
[ ] Document functional and non-functional requirements
[ ] Choose and justify the technology stack

Relevant documents and data

Project brief: LearnMate - AI assistant for smarter learning and study support..docx

Phase templates:
- Customer Journey Map (2. Requirement Analysis/Customer Journey Map.pdf)
- Data Flow Diagram (2. Requirement Analysis/Data Flow Diagram.pdf)

- Solution Requirements (2. Requirement Analysis/Solution Requirements.pdf)
- Technology Stack (2. Requirement Analysis/Technology Stack.pdf)

Project technology stack

- Node.js
- Express.js
- MongoDB
- Mongoose
- Gemini AI

Phase notes and decisions